

ERRATA FOR MEASURES FOR THE SUMMATION OF IRWIN SERIES #A11 OF VOLUME 26

Jean-François Burnol

Département de Mathématiques, Université de Lille, France
jean-francois.burnol@math.univ-lille.fr

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At the bottom of page 15, the statement of Lemma 3 is incorrect. Replace its claim

$$\mu_k(U) = b^{-l'} \mu_{k-j'}(b^{l'} U - b^{l'} x)$$

with

$$\mu_k(U) = b^{-l'} \mu_{k-j'}(b^{l'} U - b^{l'} x) + \delta_{k,j} \mathbf{1}_U(x) \frac{b}{b-1} (b^{-l} - b^{-l'}).$$

The extra term arises only if $x \in U$. For $d = 0$, the extra term vanishes: if $l' > l$, then $j' > j$, and $k \geq j'$ forces $\delta_{k,j} = 0$. For $d > 0$, the error in the proof on page 16 is in the sentence *Multiplying this by $b^{-l'}$ we obtain indeed $\mu_j(\{x\})$* . This assumes $l' = l$, apparently from a confusion with $j' = j$. For $l' > l$, when $U = \{x\}$ or, more generally, $x \in U$, the argument in the proof leads to the above extra term.

The sentence immediately preceding Lemma 3 should be modified: replacing “starting with $k = j'$ ” with “starting with $k = \max(j', j + 1)$ ” makes it true.

The only subsequent use of Lemma 3 is in the proof of Proposition 9, which follows it immediately. There, $k \geq l'$. Hence, if $l' > l$, then $k \geq l' > l \geq j$, so the problematic case $k = j$, $l' > l$ does not occur.

The paper contains a number of typographical infelicities of varying severity:

- On p. 3, in Section 2, the map from strings to integers is written as

$$n(X) = d_l b^l + \cdots + d_1 \quad \text{for } X = (d_l, \dots, d_1).$$

The leading term should of course be $d_l b^{l-1}$.

- On p. 4, in Theorem 1, in the recurrence defining $u_{j;m}$ for $j \geq 1$, the letter j is used both for the fixed first index of $u_{j;m}$ and as the summation index in

both sums. This is unfortunate; the right-hand side should read

$$\sum_{i=1}^m \binom{m}{i} \gamma_i u_{j;m-i} + \sum_{i=0}^m \binom{m}{i} d^i u_{j-1;m-i}.$$

The same problem occurs on p. 6 in Theorem 2, in the recurrence defining $v_{j;m}$, whose right-hand side should read

$$\sum_{i=1}^m \binom{m}{i} \gamma'_i v_{j;m-i} + \sum_{i=0}^m \binom{m}{i} (b-1-d)^i v_{j-1;m-i}.$$

- On p. 5, in Remark 2, the text says *Combined with the* $0 \leq u_{l;m} \leq b/(m+1)$ *bounds*. Replace l with j .
- On p. 5, Theorem 2 starts with *Let* $b > 0$. Replace it with *Let* $b > 1$.
- On p. 9, in the proof of Corollary 1, replace $\mu_{k-k(n);m+1}$ with $u_{k-k(n);m+1}$.
- On p. 15, in Remark 9, one reads that “ $\mu_k([x, x + b^{-l}])$ is 0 for $k < j$ and b^{1-l} for $j \geq k$ ”. The latter should of course read “for $k \geq j$ ”.
- In the proof of Theorem 3 at the bottom of page 16, it would have been better to write $\lim_{k \rightarrow \infty} \int_{[0,1)} \frac{\mu_{k-k(n)}(dx)}{n+x}$. The shift “ $-k(n)$ ” from Proposition 4 was deliberately omitted, as it does not affect the limit.

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